

ANDREW ALAGNA

New York, NY | 917-601-9404 | aalagna04@gmail.com | [linkedin.com/in/andrew-a-10b88215b](https://www.linkedin.com/in/andrew-a-10b88215b) | [elchic00.github.io](https://github.com/elchic00)

SOFTWARE ENGINEER | FRONTEND & FULL-STACK DEVELOPMENT | ACCESSIBILITY & PERFORMANCE

Software Engineer with 3+ years building high-performance, WCAG AAA-compliant web applications at American Express. Specialized in React.js and Node.js with proven expertise in accessibility, performance optimization, and A/B testing that delivered measurable business impact for 10M+ users across 10+ international markets.

PROFESSIONAL EXPERIENCE

American Express

Software Engineer | 08/2022 - Present

- Spearheaded Web Content Accessibility Guidelines (WCAG) AAA compliance implementation across 10+ international markets using HTML5 and React.js with axe DevTools and screen reader testing, improving accessibility audit scores from 72% to 100%, mitigating legal risk and expanding market reach for 10M+ global users
- Engineered secure user account management system using React.js and Node.js, serving 5M+ users annually to update sensitive information (name, address, contact details, income) while maintaining data integrity and regulatory compliance
- Engineered full-stack two-factor authentication (2FA) system processing 4M+ annual updates to phone numbers and email addresses, building React.js forms with real-time validation and Kotlin-based BFF API layer to handle sensitive data updates while maintaining data integrity
- Designed and executed 16+ A/B experiments using JavaScript, analytics APIs, and statistical significance testing, including A/B testing 'Find It Fast' navigation to identify and prioritize top 3 revenue-driving links, increasing form completion rates by 12% and customer satisfaction scores by 17% across 3M+ users
- Built automated CI/CD pipelines using GitHub Actions, reducing deployment time and simplifying release processes through automated version tagging, while implementing feature flagging system with real-time toggles for safer releases
- Established comprehensive testing strategy using Jest/React Testing Library in Agile environment, achieving 100% code coverage and near zero production bugs across enterprise-scale applications

CodePath

Intro to Software Engineering - Teaching Assistant | 05/2022 - 08/2022

- Led weekly instruction sessions for 180+ students in Python-based Data Structures & Algorithms, achieving 85% student satisfaction through clear explanations and engaging problem-solving
- Mentored students on technical concepts and career readiness, contributing to measurable improvements in their confidence and job search outcomes

New York City Department of Transportation
Geographic Information Systems Intern | 04/2022 - 08/2022

- Independently designed and built Python automation to process annual census data and integrate with ArcGIS Pro, reducing manual processing time from weeks to seconds and enabling data-driven policy decisions through district-level demographic maps
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VOLUNTEER LEADERSHIP

CodePath | 06/2021 - Present

Teaching & Mentoring

- Currently leading weekly mentorship sessions for 5 students in open source contribution course, teaching GitHub workflows, PR creation, and early career preparation with async support
- Mentored and instructed 350+ students across multiple cohorts in Data Structures & Algorithms, cybersecurity fundamentals, and career readiness, fostering an inclusive learning environment
- Taught technical curriculum and guided students through hands-on debugging and technical interview preparation, securing free certifications for 300+ students

CUNY: Hunter College Alumni Mentorship Program | 2025 - Present

Teaching & Mentoring

- Providing 1-on-1 career mentorship to 2 current students on side project ideation, resume optimization, technical interviews, and navigating the tech job market through video sessions and LinkedIn messaging
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EDUCATION

Bachelor of Arts in Computer Science

Hunter College | New York, NY | Graduated: May 2022 | GPA: 3.62, Cum Laude

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL, Kotlin

Frontend: React.js, Next.js, React Context API, Redux, HTML5, CSS3, Tailwind CSS, Responsive Design, Component Libraries, Single Page Applications

Backend & APIs: Node.js, GraphQL, REST APIs, BFF Architecture, Microservices

Tools & Practices: Git, GitHub Actions (CI/CD), Jest, React Testing Library, WCAG AAA Accessibility, Agile/Scrum, A/B Testing, Performance Optimization, AWS, Docker, Jenkins